L2.5 Curve Sketching

§3.5 — precalc + the derivatives table

The two halves — run them in this order

① **Precalc** — ***Buenos días***	② **Calculus**
D Domain	f' $\Rightarrow$ f $\nearrow\searrow$
I Intercepts, and the *type* of each zero	CNs, and which are extrema
A Asymptotes: VA, HA, SA — or **EB**	f'' $\frown\smile$
S Symmetry	CNs *and* PIPs, then the sign check

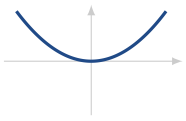
THE DERIVATIVES TABLE — one number line, five steps

① Find the **CNs** and the **PIPs**.
② Mark them all on **one** number line.
③ Write the **evaluated** sign of f' and f'' on each interval.
④ **Draw it:** arrows above the line, cups below it.
⑤ **Sketch** f underneath, at the same horizontal positions.

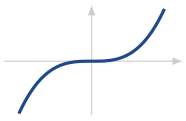
CN: $f'(c) = 0$ *or* $f'(c)$ DNE PIP: $f''(c) = 0$ *or* $f''(c)$ DNE

ZERO AND VA TYPES — read them off the FACTORED form

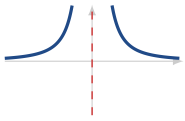
B	**bounce**	even power on the factor — touches the axis and turns back
PT	**pass through**	odd power — crosses the axis
	symmetric VA	even power on a bottom factor — both sides go the same way
	flip VA	odd power on a bottom factor — the sides go opposite ways



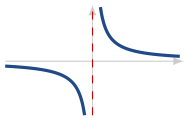
bounce: $y = x^2$



pass through: $y = x^3$

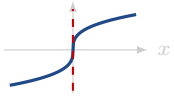


symmetric: $y = \dfrac{1}{x^2}$



flip: $y = \dfrac{1}{x}$

VERTICAL TANGENTS



Where f' **fails to exist** but the point is still in the domain, the curve has a **vertical tangent** — and it is drawn and labeled on the sketch like any extremum.

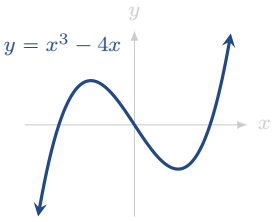
Ex. $y = \sqrt[3]{x}$: $y' = \dfrac{1}{3x^{2/3}}$ does not exist at $x = 0$, which is in the domain, so there is a vertical tangent at $(0, 0)$.

END BEHAVIOR, HA AND SA

EB is how the function behaves as $x \to +\infty$ and as $x \to -\infty$. A rational function has an **HA** when the degree on the bottom is higher or the degrees are equal, and an **SA** when the degree on top is higher by one. For the SA, long division gives $f(x) = mx + b + R(x)$ with $R(x) \to 0$, so $y = mx + b$ is the slant asymptote; draw it dashed on the axes. A polynomial has neither, so write its **EB**.

Ex. $f(x) = x^3 - 4x$. The leading term x^3 has a positive coefficient and odd degree, so

$$\begin{aligned} x \to -\infty &\;\Rightarrow\; f(x) \to -\infty \\ x \to +\infty &\;\Rightarrow\; f(x) \to +\infty \end{aligned}$$



WORKED EXAMPLE — $f(x) = \sqrt[3]{x}\,(x-4)$

D all real numbers
I zeroes $x = 0$ (PT), $x = 4$ (PT); y-int 0
A no VA, HA or SA; **EB**: $f \to +\infty$ both ends
S $f(-x) = \sqrt[3]{x}\,(x+4)$ — neither

$$f(x) = x^{4/3} - 4x^{1/3}$$

$$f'(x) = \frac{4(x-1)}{3x^{2/3}} \quad \text{CNs: } 0 \text{ (DNE)}, 1$$

$$f''(x) = \frac{4(x+2)}{9x^{5/3}} \quad \text{PIPs: } -2,\ 0$$

